№2

2.1

□ Пусть $P(0)=p_1$, $P(1)=p_2$

$Q(0)=q_1$, $Q(1)=q_2$

$f(0)=a_1$ $f(1)=$

$$a_1^2\cdot\frac{p_1^2}{q_1}+a_2^2\cdot\frac{p_2^2}{q_2}<a_1^2p_1+a_2^2p_2$$

Возьмем $a_1=10$, $a_2=0$, $p_1=0{,}1$, $q_1=0{,}9$. Имеем:

$$100\cdot\frac{0{,}01}{0{,}9}\wedge 100\cdot 0{,}1 \Rightarrow 1\wedge 100\cdot 0{,}1\cdot 0{,}9=9\Rightarrow 1<9$$

■

2.2

П В силу марковости:

$$p_\pi(\tau)=\prod_{t=0}^{T}\pi(a_t|s_t)\cdot P_t(s_{t+1}|s_t,a_t)$$

Аналогично можно написать для p_μ.

$$\mathbb{E}_\pi\frac{p_\pi(\tau)}{p_\mu(\tau)}=\sum_\tau p_\pi(\tau)\cdot\frac{p_\pi(\tau)}{p_\mu(\tau)}$$

Возьмем логарифм, и воспользуемся нер-ом Йенсена:

$$\log\mathbb{E}_\pi\frac{p_\pi(\tau)}{p_\mu(\tau)}\geqslant\mathbb{E}_\pi\log\frac{p_\pi(\tau)}{p_\mu(\tau)}=\mathbb{E}_\pi\sum_{t=1}^{T}\log\frac{\pi(a_t|s_t)\,P(s_{t+1}|s_t,a_t)}{\mu(a_t|s_t)\cdot P(s_{t+1}|s_t,a_t)}=$$

$$=\sum_{t=1}^{T}\mathbb{E}_\pi\log\frac{\pi(a_t|s_t)}{\mu(a_t|s_t)}=\sum_{t=1}^{T}\mathbb{E}_{s_t\sim p_t^\pi}\left[KL(\pi(\cdot|s_t)|\mu(\cdot|s_t))\right]\Rightarrow$$

$$\Rightarrow\mathbb{E}_\pi\frac{p_\pi(\tau)}{p_\mu(\tau)}\geqslant\prod_{t=1}^{T}\mathbb{E}_{s_t\sim p_t^\pi}\left[KL(\pi(\cdot|s_t)|\mu(\cdot|s_t))\right]$$

■